HELIOS CANOPY

Staged Commissioning Guide

Solar Thermal Greenhouse Heating System

Southwest Finland

Feature: 017-review-hardware-architecture

Date: 2026-04-02

This guide describes how to bring the greenhouse solar heating control system online incrementally as hardware installation progresses. You enable individual loops as hardware is installed, with hard safety rules (freeze/overheat drain) always active from the first power-on.

Generated: 2026-04-04

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Overview Flowchart

The commissioning process follows four phases. See the full Mermaid diagram at `design/diagrams/commissioning-flow.mmd`.

```
┌──────────────────┐     ┌──────────────────┐     ┌──────────────────┐     ┌──────────────────
│  Phase 1         │     │  Phase 2         │     │  Phase 3         │     │  Phase 4
│  Install         │────►│  Verify          │────►│  Solar Loop      │────►│  Add Loops
│  Control Box     │     │  Sensors         │     │  (Stage 1)       │     │  Progressively
│                  │     │                  │     │                  │     │
│  • Pro 4PM       │     │  • Read sensors  │     │  • ce=true       │     │  • Add GH loop
│  • 3 sensors     │     │  • Warm/cool     │     │  • ea=3 (V+P)    │     │  • Add EH mode
│  • Pro 2PMs      │     │    each probe    │     │  • am=["SC"]     │     │  • am=null (all)
│  • 24V PSU       │     │  • Confirm IDs   │     │  • Test freeze   │     │  • ea=31 (all)
│  • Ethernet      │     │                  │     │  • Autonomous    │     │  • Full auto
└──────────────────┘     └──────────────────┘     └──────────────────┘     └──────────────────

                                                        |
                                                   Safety drains
                                                 ALWAYS active ⚡
```

Safety guarantee: Freeze drain (outdoor < 2°C) and overheat drain (tank > 85°C) fire automatically at every stage, regardless of device config settings. They cannot be disabled.

Prerequisites

Before starting commissioning:

- [ ] Shelly Pro 4PM flashed with latest scripts via `shelly/deploy.sh`
- [ ] Ethernet switch installed and all Pro devices connected
- [ ] 24V DC PSU wired to all valve actuators
- [ ] Shelly 1 Gen3 + Add-on connected with 3 DS18B20 sensors
- [ ] Router configured with DHCP reservations matching `shelly/devices.conf`

Phase 1: Install Control Box

Hardware Checklist

Component	IP Address	Purpose
Shelly Pro 4PM	192.168.1.174	Controller — runs control logic, drives pump
Shelly Pro 2PM #1	192.168.1.11	Valves VI-btm, VI-top (input manifold)
Shelly Pro 2PM #2	192.168.1.12	Valves VI-coll, VO-coll (collector loop)
Shelly Pro 2PM #4	192.168.1.14	Valves V_ret, V_air (collector top)
Shelly 1 Gen3 + Add-on	192.168.1.86	Sensor hub — 3× DS18B20
Zyxel GS-108BV5	—	8-port Ethernet switch
Mean Well 24V 15W PSU	—	Powers all valve actuators

Step 1.1: Deploy Scripts

```
cd shelly/
./deploy.sh
```

This flashes `control-logic.js` + `control.js` to the Pro 4PM (script slot 1) and `telemetry.js` (slot 3).

Step 1.2: Initial Boot (Controls Disabled)

The system boots with controls disabled by default (`ce: false`). Push this initial config via the Device view in the playground UI or via MQTT:

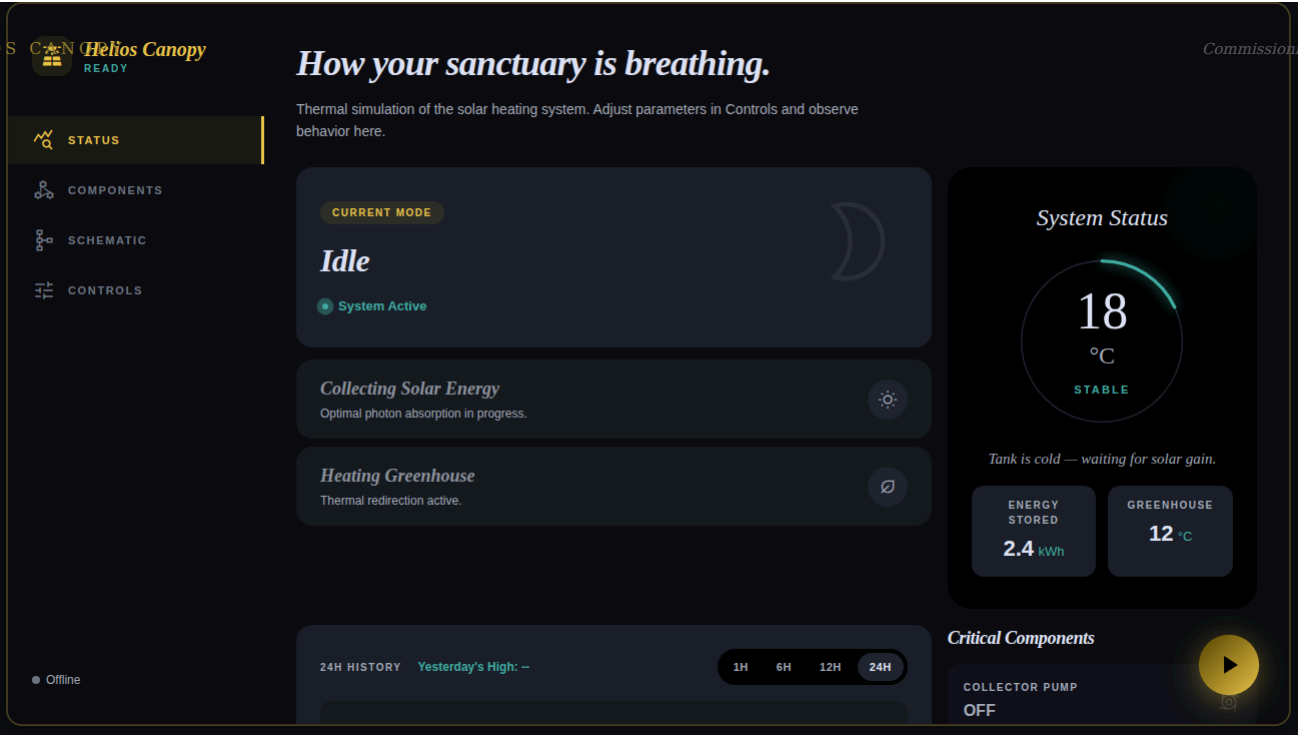
```
{ "ce": false, "ea": 0, "fm": null, "am": null, "v": 1 }
```

At this point:

- The system monitors all sensor temperatures
- No valves or actuators are commanded
- **Safety drains are still active** — if outdoor drops below 2°C, the system WILL drain the collectors even with `ce: false`

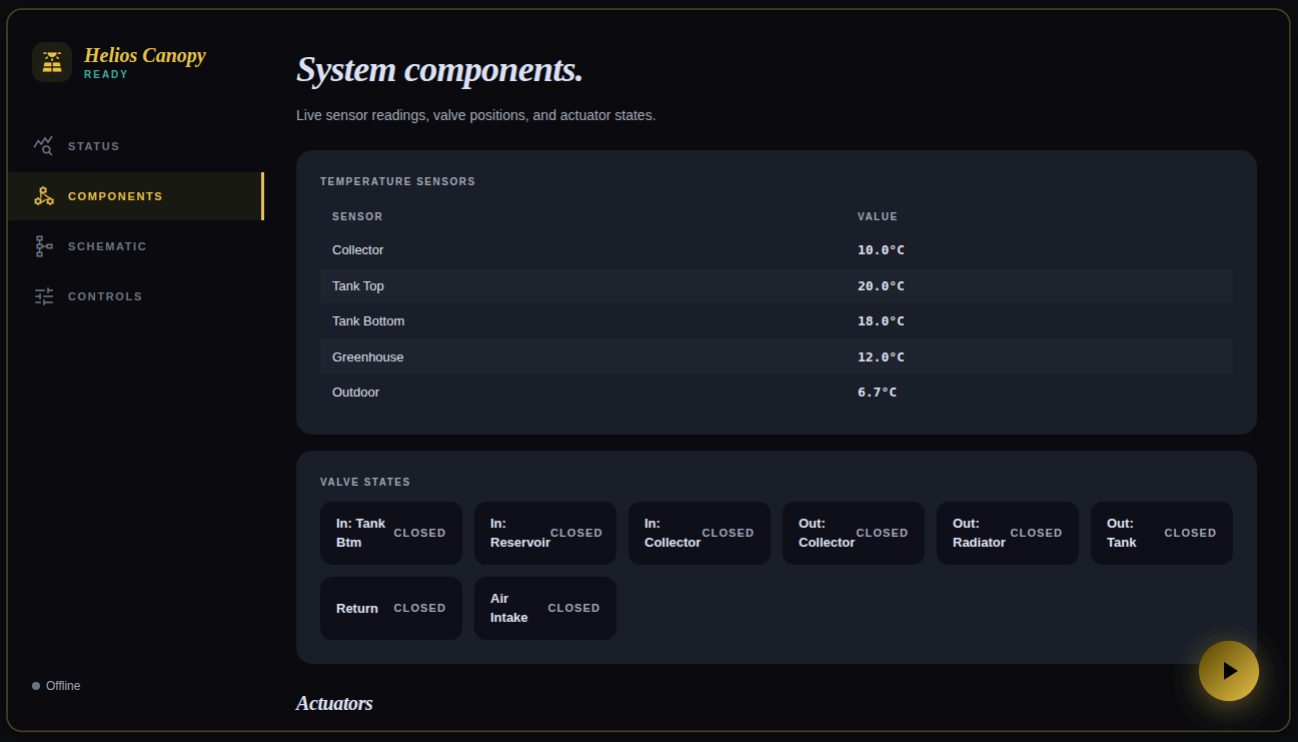
Step 1.3: Verify Communication

Open the playground UI (connect to your server or use the live mode toggle). Navigate to **Status** view.



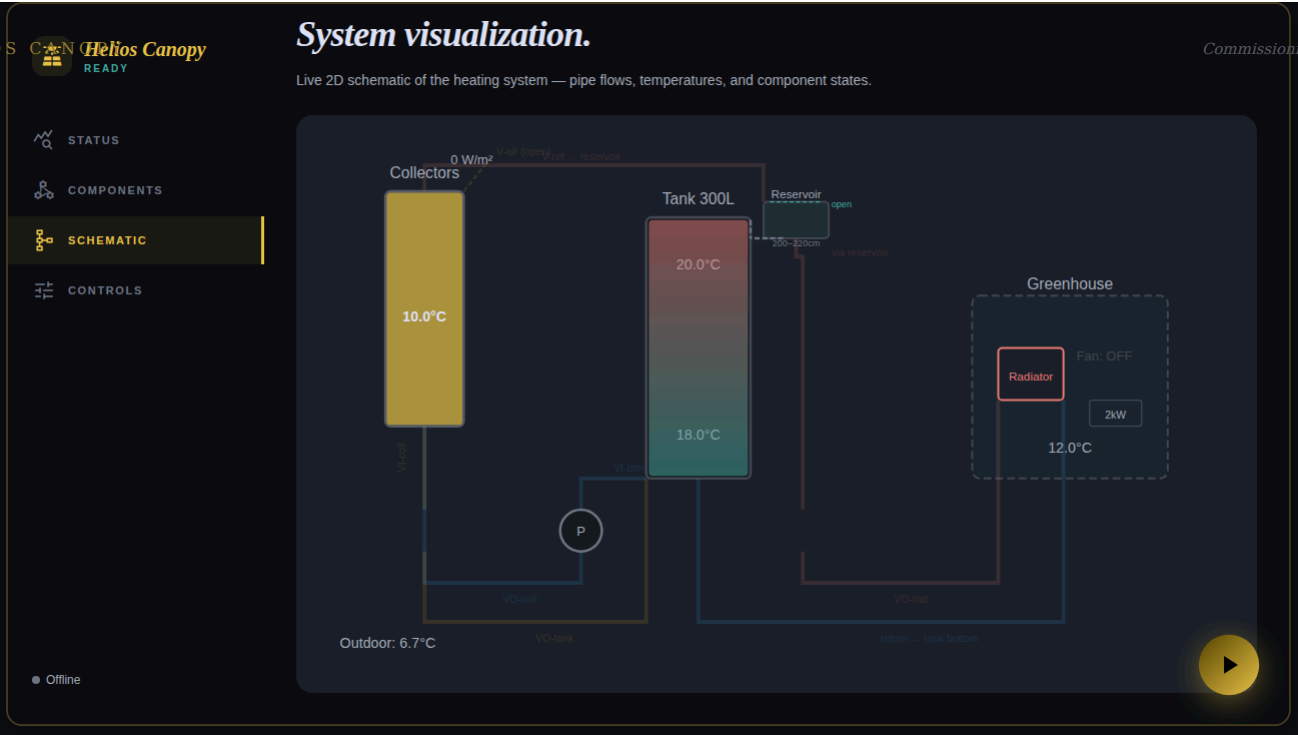
Status view after first boot. The mode shows "Idle", and sensor readings populate in the gauge and cards. All sensors should show plausible temperatures.

Navigate to **Components** view to see individual sensor readings and valve states:



Components view showing all temperature sensors with current readings, all valves CLOSED, and all actuators OFF.

Navigate to **Schematic** view to see the system topology:



Schematic view showing the physical layout. All pipe flow paths are dimmed (inactive). The pump circle shows no activity. Temperatures are displayed at each component.

Phase 2: Sensor Identity Verification

Why this matters: DS18B20 sensors on the 1-Wire bus are identified by index (0-4), not by physical label. A miswired sensor — say the outdoor sensor probe is actually reading collector temperature — could prevent freeze drain from triggering, or trigger it falsely. **You must verify each sensor identity before enabling automatic modes.**

Step 2.1: Read Current Sensor Values

With the system booted (`ce: false`), open the playground UI → **Components** view. You should see 3 temperature readings:

Sensor Name	Expected Location	Expected Range
Collector	Collector outlet (280cm height)	Ambient or sun-warmed
Tank Bottom	Tank lower region (~10cm)	Water temperature
Outdoor	Outside, shaded	Outside air temperature

Step 2.2: Physical Verification (One Sensor at a Time)

For each of the 3 sensors:

1. **Locate the physical probe** on the 1-Wire bus
2. **Apply heat** — grip the probe firmly with your hand (body heat is enough for a 2-3°C rise), or use a heat gun / warm water
3. **Watch the playground UI** — the Components view updates every 30 seconds. One reading should rise noticeably
4. **Confirm the correct reading changed**:
    - If you warmed the probe at the collector outlet and the "Collector" reading rose → correct
    - If a different reading rose → the sensor wires are swapped

Step 2.3: Fix Miswiring

If sensors are swapped:

- **Option A**: Physically swap the sensor probes to the correct locations
- **Option B**: Swap the sensor IDs in `control.js` → `SENSOR_IDS` map → redeploy

Step 2.4: Freeze Trigger Test

Use ice wrapped in cloth on the **outdoor sensor** probe:

1. Watch the outdoor temperature reading drop toward 2°C

2. When it drops below 2°C, the system should automatically enter **Active Drain** mode
3. Verify mode change in the Status view (mode badge changes to "Active Drain")
4. Remove the ice — outdoor reading recovers, system returns to Idle

> **Note**: This test fires even with `ce: false` because freeze drain is a hard safety override.

Step 2.5: Record Verified Mapping

Wire Label	Physical Location	Logical Name	Sensor ID	Verified
_____	Collector outlet	collector	0	☐
_____	Tank lower region	tank_bottom	2	☐
_____	Outside, shaded	outdoor	4	☐

Phase 3: Enable Solar Collector Loop (Stage 1)

Step 3.1: Set Device Config

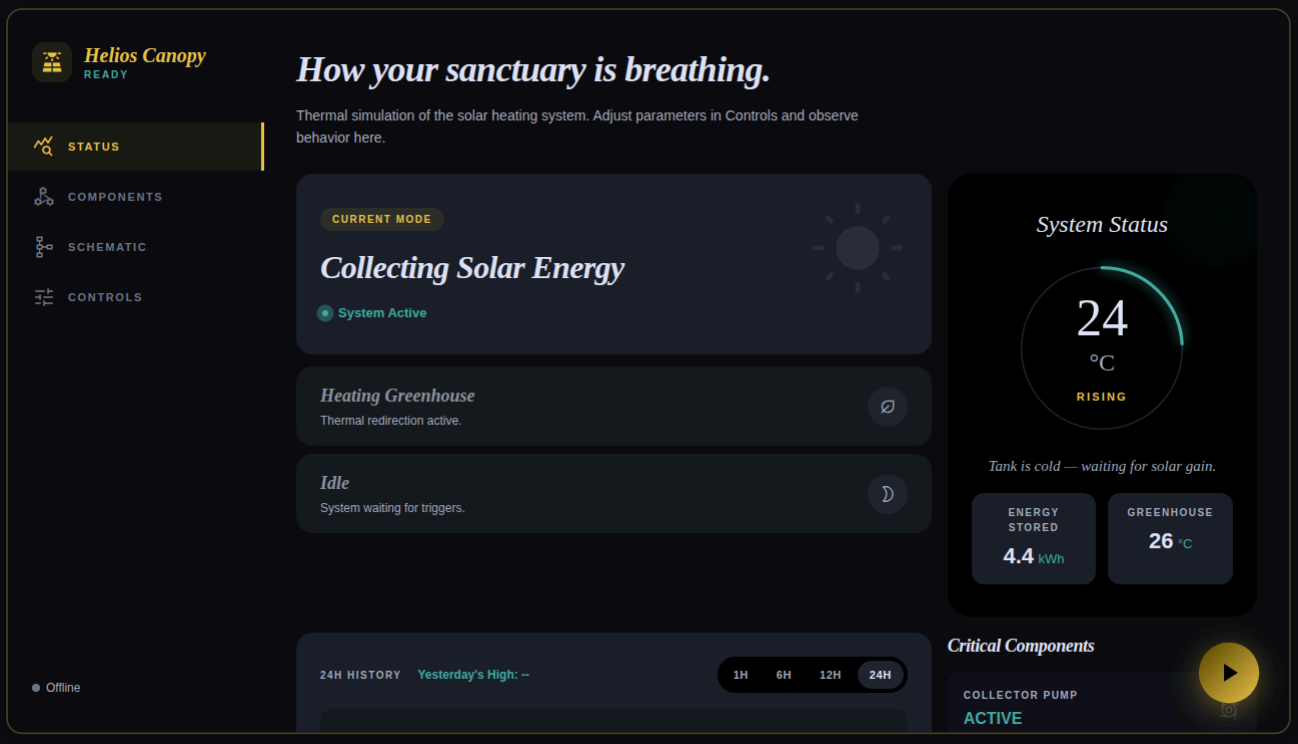
Push this config via the playground Device view or MQTT:

```
{ "ce": true, "ea": 3, "fm": null, "am": ["SC"], "v": 2 }
```

Field	Value	Meaning
`ce`	`true`	Controls enabled — system can actuate valves and pump
`ea`	`3`	Enabled actuators: valves (bit 0 = 1) + pump (bit 1 = 2)
`am`	`["SC"]`	Only Solar Charging mode allowed. Safety drains always bypass this filter.

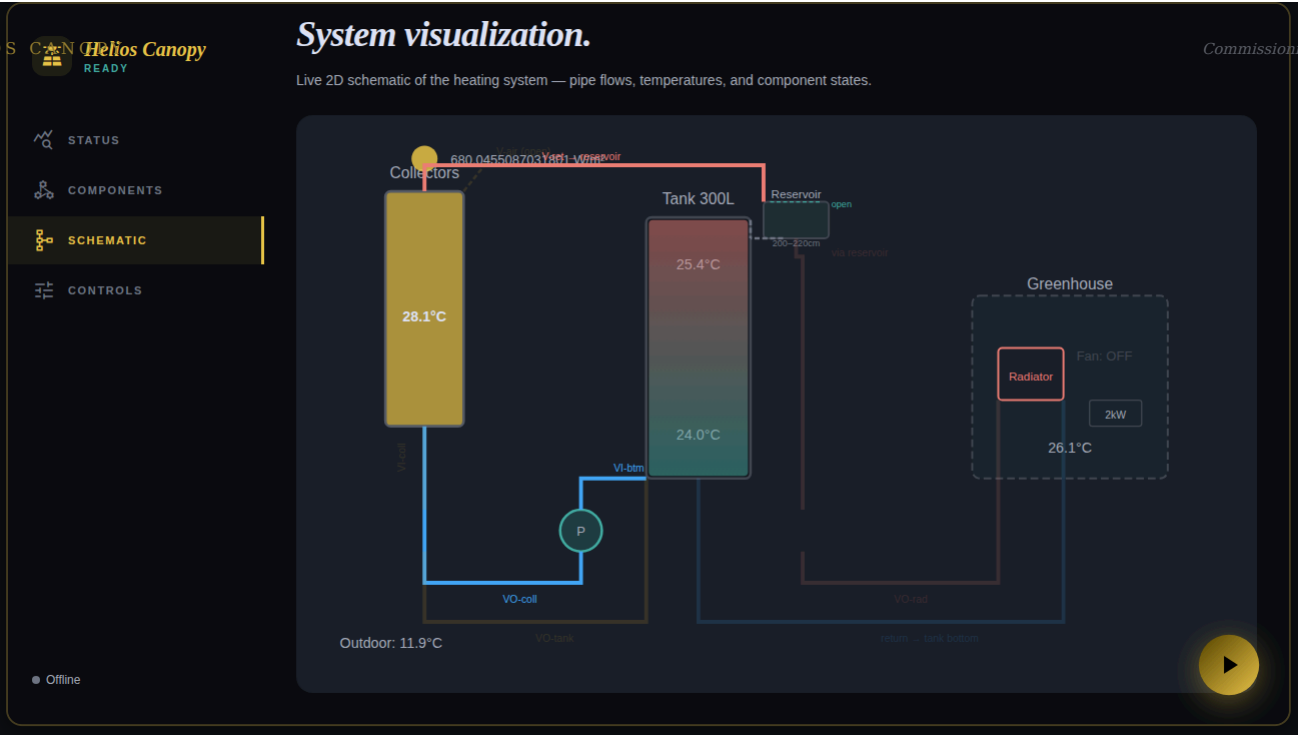
Step 3.2: Verify Solar Charging

Wait for sunny conditions (collector temperature rises above tank bottom + 7°C). The system should automatically enter Solar Charging mode.



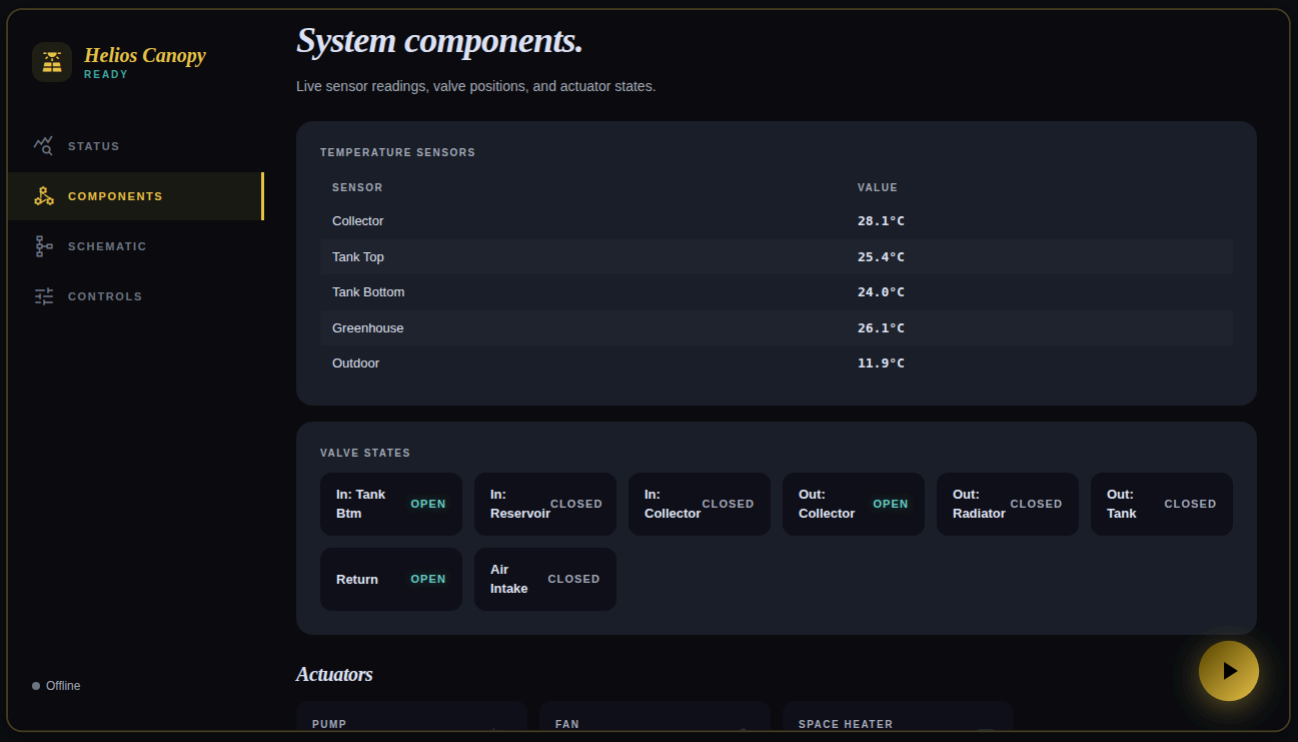
Status view during Solar Charging. The mode badge shows "Collecting Solar Energy" with a system active indicator. The tank gauge shows stored energy rising.

The **Schematic** view shows the active flow path:



Schematic during solar charging. The solar collector pipes are highlighted, showing water flow from tank bottom → pump → collectors → reservoir → tank top. The pump circle pulses to show it is running.

The **Components** view confirms which valves are open:



Components view during solar charging. Valves VI-btm, VO-coll, and V_ret show OPEN. All other valves are CLOSED. The pump shows ACTIVE.

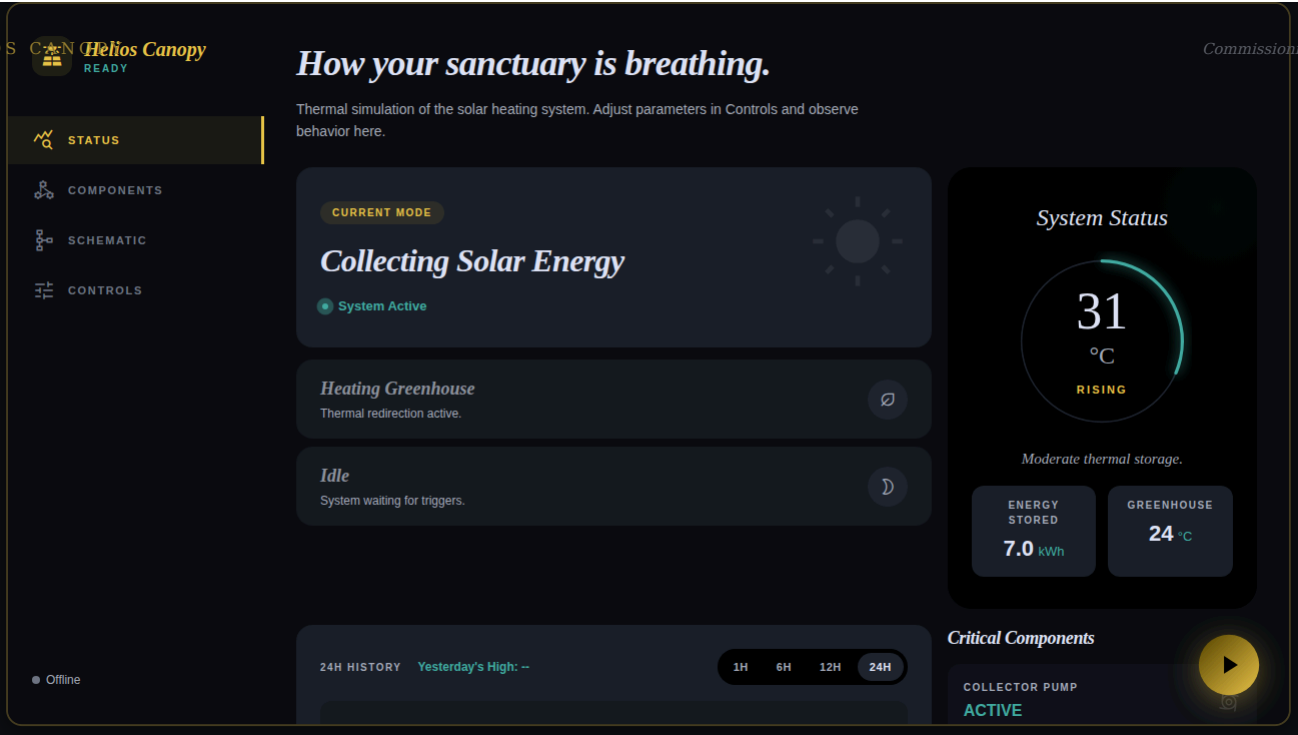
Step 3.3: Manual Freeze Drain Test

Even if outdoor temperature is well above 2°C, you can manually trigger a drain to verify the full sequence:

1. Push forced mode config:

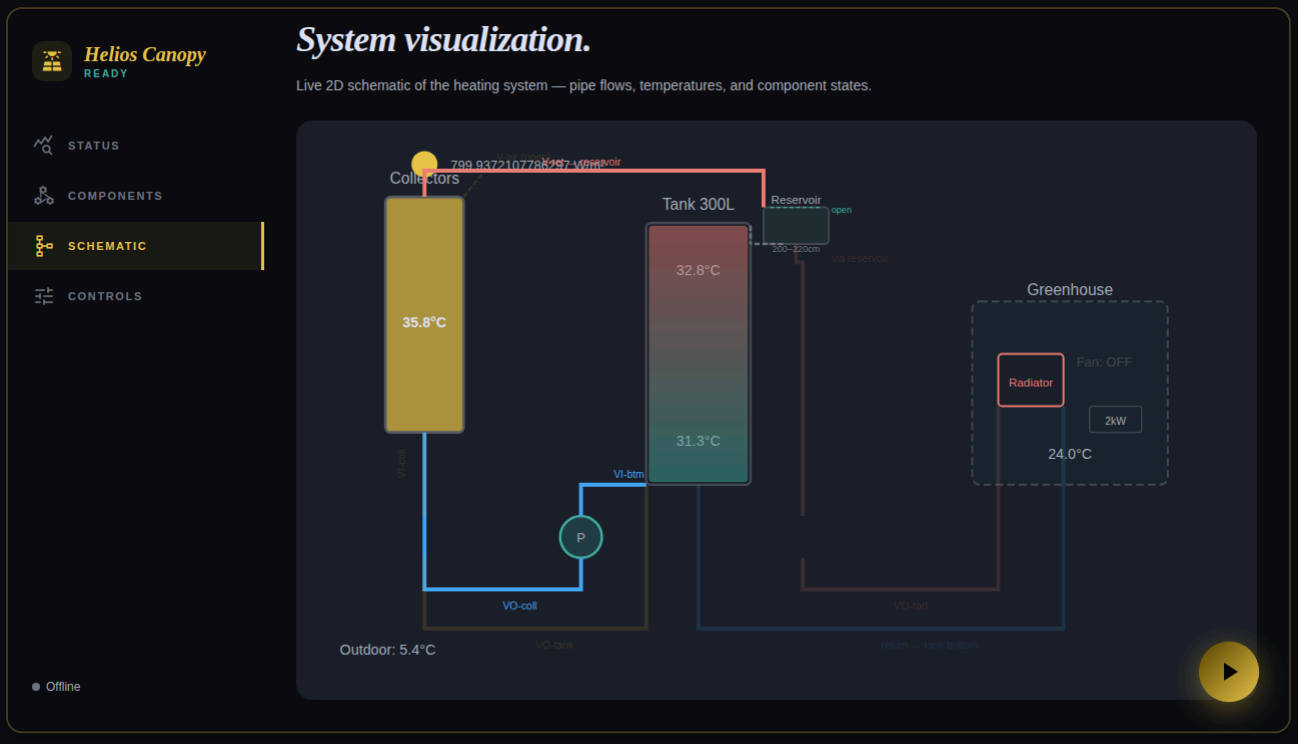
   ```json
   { "ce": true, "ea": 3, "fm": "AD", "am": ["SC"], "v": 3 }
   ```

1. The system immediately enters Active Drain mode. Watch the transition:



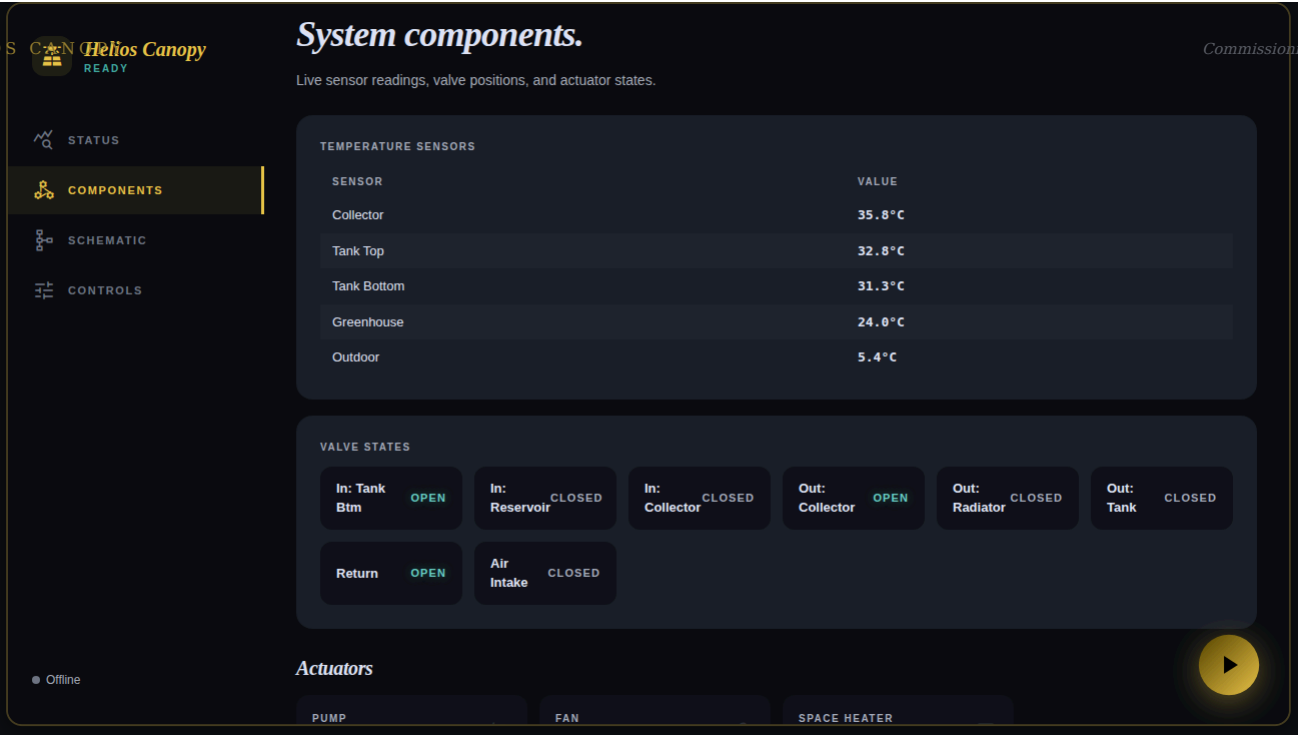
Status view during Active Drain (freeze protection). The mode is shown as active with drain-specific indicators.

The schematic shows the drain flow path:



Schematic during Active Drain. The drain pipes are highlighted: V_air opens at collector top (air intake), pump draws water from collector bottom (VI-coll) and pushes to tank (VO-tank).

The components view confirms the drain valve configuration:



Components view during Active Drain. Valves VI-coll, VO-tank, and V_air show OPEN. Pump is ACTIVE. All other valves are CLOSED.

1. The drain sequence runs until:
   - Pump power drops below 20W for 3 consecutive readings (600ms) → dry-run detected → pump stops
   - Or 180-second timeout → pump stops
1. Clear forced mode:

```json
{ "ce": true, "ea": 3, "fm": null, "am": ["SC"], "v": 4 }
```

Step 3.4: Stage 1 Validation Checklist

- ☐ All 3 sensors reading plausible temperatures (Components view)
- ☐ Solar charging activates automatically when collector is hot (collector > tank_bottom + 7°C)
- ☐ Solar charging stops when delta drops below 3°C
- ☐ Freeze drain fires automatically when outdoor < 2°C
- ☐ Manual drain test (`fm: "AD"`) executes full sequence
- ☐ Pump dry-run detection stops drain correctly (power monitoring)
- ☐ Collectors marked as drained after drain completes
- ☐ System returns to Idle after drain, resumes solar when conditions are met

Stage 1 is now operational. The solar collector loop runs autonomously with freeze protection. No other modes can activate (greenhouse heating, emergency heating are filtered by `am`).

Phase 4: Progressive Enablement

As you install additional hardware (radiator loop, space heater), progressively enable modes. At each step: verify new sensors, test the new mode manually via forced mode, then add it to the allowed modes list.

Add Greenhouse Heating

When: Radiator + fan installed in greenhouse, tank top sensor and greenhouse sensor connected.

New hardware:

- Pro 2PM unit 3 (192.168.1.13) — valves VO-rad, VO-tank
- Tank top sensor (sensor ID 1) and greenhouse sensor (sensor ID 3)
- 230V fan connected to Pro 4PM O2

Sensor verification: Repeat Phase 2 for the 2 new sensors (tank_top, greenhouse). Warm each probe and verify the correct reading changes in Components view.

Config update:

```
{ "ce": true, "ea": 7, "fm": null, "am": ["SC", "GH"], "v": 5 }
```

Change	Old → New	Meaning
`ea`	`3 → 7`	Added fan (bit 2 = 4): valves + pump + fan
`am`	`["SC"] → ["SC", "GH"]`	Added Greenhouse Heating mode

Test: Set `fm: "GH"` to force greenhouse heating. Verify:

- Pump runs, fan runs
- Hot water flows through radiator (VI-top → pump → VO-rad → radiator → tank bottom)
- Greenhouse temperature begins to rise
- Clear `fm: null` when verified

Validation:

- ☐ Tank top and greenhouse sensors verified
- ☐ Forced greenhouse heating (`fm: "GH"`) operates pump + fan
- ☐ Radiator loop circulates (feel pipes warming)
- ☐ System auto-enters GH mode when greenhouse < 10°C and tank has useful heat

Add Emergency Heating

When: 2kW space heater connected to Pro 4PM O4.

Config update:

```
{ "ce": true, "ea": 15, "fm": null, "am": ["SC", "GH", "EH"], "v": 6 }
```

Change	Old → New	Meaning
`ea`	`7 → 15`	Added space heater (bit 3 = 8)
`am`	`["SC", "GH"] → ["SC", "GH", "EH"]`	Added Emergency Heating mode

Test: Set `fm: "EH"` to force emergency heating. Verify space heater activates (feel heat output). Clear `fm: null`.

Full System

When: All hardware installed and each loop individually verified.

Config update:

```
{ "ce": true, "ea": 31, "fm": null, "am": null, "v": 7 }
```

Change	Old → New	Meaning
`ea`	`15 → 31`	Added immersion heater (bit 4 = 16): all actuators enabled
`am`	`["SC", "GH", "EH"] → null`	All modes allowed (no filter)

The system now operates fully autonomously. All modes can activate based on sensor conditions. Safety drains have been active since Phase 3.

Device Config Quick Reference

Actuator Bitmask (`ea`)

Bit	Value	Actuator	Added at
0	1	Motorized valves	Stage 1
1	2	Circulation pump	Stage 1
2	4	Radiator fan	Greenhouse heating
3	8	Space heater (2kW)	Emergency heating
4	16	Immersion heater	Full system

Common combinations: `3` (V+P), `7` (+fan), `15` (+heater), `31` (all).

Mode Codes (`fm` , `am`)

Code	Mode	Valves Open	Actuators
`I`	Idle	None	None
`SC`	Solar Charging	VI-btm, VO-coll, V_ret	Pump
`GH`	Greenhouse Heating	VI-top, VO-rad	Pump, Fan
`AD`	Active Drain	VI-coll, VO-tank, V_air	Pump
`EH`	Emergency Heating	None	Space heater, Immersion

Safety Rules (Always Active)

These fire regardless of `ce` , `ea` , or `am` :

Rule	Trigger	Action
Freeze drain	outdoor < 2°C	Active Drain (empties collectors)
Overheat drain	tank_top > 85°C	Active Drain (prevents boiling)
Sensor staleness	Any sensor > 150s old	Idle (pump off, safe state)

Troubleshooting

Sensor reading shows "--" or null

- Check physical connection of DS18B20 probe to Add-on
- Verify sensor hub IP (192.168.1.86) is reachable from Pro 4PM
- Check 1-Wire bus wiring (data, VCC, GND)

Valve doesn't open

- Verify 24V PSU is powered and wired to valve actuators
- Check Pro 2PM unit IP is reachable from Pro 4PM
- Check valve relay assignment in `control.js` VALVES map
- Test directly: access Pro 2PM web UI → Switch → toggle relay

Pump doesn't start

- Verify pump is connected to Pro 4PM O1 (id: 0)
- Check `ea` bitmask includes pump (bit 1 = 2)
- Check `ce: true` in device config
- For safety drains: pump starts even with `ce: false`

Solar charging doesn't activate

- Check `am` includes `"SC"` (or is `null`)
- Verify collector temp > tank_bottom + 7°C (check Components view)
- If collectors are drained, system waits for speculative refill conditions
- Check `collectorsDrained` flag in system logs

Freeze drain doesn't fire

- Verify outdoor sensor is correctly identified (Phase 2)
- Check outdoor reading is actually below 2°C (Components view)
- Safety drain bypasses `ce`, `ea`, `am` — if it still doesn't fire, check sensor hub connectivity